

TAOLAGNARO, Madagascar

WMO: 671970

Lat: 25.038S

Lon: 46.956E

Elev: 29

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	59.1	61.0	51.7	56.9	68.1	53.6	61.2	67.4	27.6	73.3	25.3	73.2	2.5	320	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	89.9	78.3	88.9	78.0	87.7	77.7	80.2	86.6	79.5	85.7	78.9	84.8	16.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	149.0	84.4	77.4	142.8	83.0	77.1	141.2	82.7	43.9	87.2	43.1	86.0	42.4	85.5	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
29.8	26.9	24.0	DB	55.2	93.7	1.7	1.9	53.9	95.1	53.0	96.2	52.0	97.3	50.8	98.7
			WB	53.2	82.8	1.6	1.5	52.0	83.8	51.1	84.7	50.2	85.5	49.0	86.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.6	80.8	80.7	79.7	77.1	73.4	70.8	69.5	70.8	72.6	75.0	77.0	79.8
	DBStd	4.88	2.56	2.36	2.54	3.04	2.77	2.68	2.74	2.65	3.00	3.21	3.21	3.00
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDD50	9335	956	859	922	812	725	624	606	644	678	775	809	925
	CDD65	3862	491	439	457	362	260	174	143	179	228	310	359	460
	CDH74	26729	4669	4177	3960	2401	1040	489	314	552	975	1684	2402	4066
	CDH80	6686	1408	1221	1039	485	101	27	9	38	130	352	628	1248
(14) Wind	WSAvg	11.4	12.0	11.5	11.1	9.6	8.8	8.5	9.4	11.9	13.9	14.0	13.7	12.6
(15) Precipitation	PrecAvg	64.6	9.3	8.5	7.9	6.4	5.4	3.4	4.7	3.6	1.5	3.3	4.3	6.4
	PrecMax	101.5	39.2	25.5	29.6	16.7	13.5	7.2	20.4	9.3	4.0	10.6	19.6	16.0
	PrecMin	41.5	2.2	0.6	0.7	0.9	0.6	0.0	0.2	0.0	0.1	0.2	0.0	1.3
	PrecStd	15.7	8.4	6.5	6.3	3.9	3.4	2.1	4.1	2.4	1.1	2.8	4.0	4.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	91.1	90.1	88.2	84.5	82.6	80.9	82.8	85.1	87.8	89.6	91.3
		MCWB	79.6	79.0	78.6	77.4	74.2	71.5	69.5	70.2	72.9	75.4	76.6	78.4
	2%	DB	89.7	89.3	88.2	86.3	82.5	80.5	78.9	80.8	82.8	85.6	87.5	89.5
		MCWB	78.5	78.2	78.1	76.4	73.4	70.8	68.9	69.8	71.3	74.4	75.9	78.1
	5%	DB	88.1	87.8	86.8	84.5	80.7	78.7	77.2	79.0	81.3	83.8	85.8	87.9
		MCWB	78.2	78.0	77.6	75.6	72.5	69.8	68.2	69.1	70.5	73.2	75.2	77.6
	10%	DB	86.4	86.2	85.6	82.7	79.0	76.9	75.5	77.2	79.5	82.1	84.0	86.2
		MCWB	77.6	77.6	77.1	74.8	71.7	68.9	67.6	68.4	69.8	72.5	74.4	77.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	81.5	80.6	79.4	76.7	73.7	72.0	73.0	74.7	77.6	79.1	80.7
		MCDB	88.3	87.1	87.3	85.8	81.6	79.5	77.9	79.5	82.6	85.4	86.5	88.2
	2%	WB	80.1	80.0	79.5	78.0	75.2	72.3	70.6	71.6	72.9	75.6	77.5	79.4
		MCDB	86.3	85.9	85.5	83.9	80.0	77.8	76.3	78.2	80.0	82.6	84.8	86.4
	5%	WB	79.3	79.2	78.6	77.0	73.9	71.1	69.5	70.3	71.9	74.4	76.3	78.6
		MCDB	85.2	84.9	84.3	82.5	78.8	76.4	75.0	76.4	78.9	80.9	83.2	85.2
	10%	WB	78.6	78.5	77.9	76.0	72.7	70.0	68.7	69.3	70.9	73.3	75.2	77.8
		MCDB	84.3	83.9	83.3	81.2	77.6	75.1	74.0	75.1	77.4	79.5	81.5	84.3
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.0	10.8	11.3	11.8	12.5	12.3	13.1	13.2	12.5	12.0	11.7
		MCDBR	13.2	13.0	12.5	13.0	13.7	15.1	14.9	16.1	16.0	15.2	14.9	13.9
		MCWBR	6.0	5.9	5.9	6.5	7.5	8.5	8.3	8.9	8.4	7.5	7.2	6.3
	5% WB	MCDBR	12.0	11.6	11.5	12.2	12.3	13.7	13.9	14.9	15.1	14.1	13.8	13.0
		MCWBR	6.1	6.0	5.9	6.7	7.6	8.3	8.6	9.0	8.6	7.5	7.5	6.6
(40) Clear-Sky Solar Irradiance	taub		0.421	0.419	0.409	0.386	0.361	0.344	0.344	0.364	0.409	0.459	0.436	0.425
	taud		2.465	2.467	2.496	2.541	2.594	2.605	2.593	2.509	2.358	2.247	2.358	2.431
	Ebn at Noon		293	289	283	276	270	269	273	278	278	274	287	292
	Edh at Noon		38	37	34	30	26	25	26	31	39	46	42	39
(44) All-Sky Solar Radiation	RadAvg		2161	2031	1813	1503	1249	1120	1175	1464	1815	2001	2150	2248
	RadStd		156	136	119	69	43	26	34	70	69	117	108	108

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.36	N/A	+1.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page